doesn’t see them shut their browser down while waiting forever to display a page as the server processes the request. The most common processing bottleneck in a dynamic web application exists where the application communicates with the database. You will learn how to use caching to prevent unnecessary database operations when working with data that doesn’t change very often. You will also look at using the cache tag helper to prevent unnecessary regeneration of sections of dynamic UI.

Finally, you will put your application into production. You will review some of the hosting options available to you and learn how to configure publishing your application to target some of those different options. Then you will focus on the steps required to deploy to Internet Information Services (IIS), Microsoft’s primary web server in a Windows environment.

14.1 Working with environments

In chapter 9, you created a working implementation of the `EmailSender` interface, so you can test sending email confirmations to users when they register an account with your application. If you recall, the implementation that you developed writes emails to a folder on disk. This is very useful during the development phase because it enables you to easily access and read the emails you generate without having to rely on network connectivity. As you can imagine, this implementation is of limited use for an application when it is live. You will need another implementation that connects to a mail server and sends email properly. Once you have that, you need to be able to switch between the two implementations, depending on whether you are running the application in production or developing it further. The solution to this problem is a feature called *environments*. You have encountered this feature on a number of occasions previously. Now you will take a closer look at what it is and how you can use it.